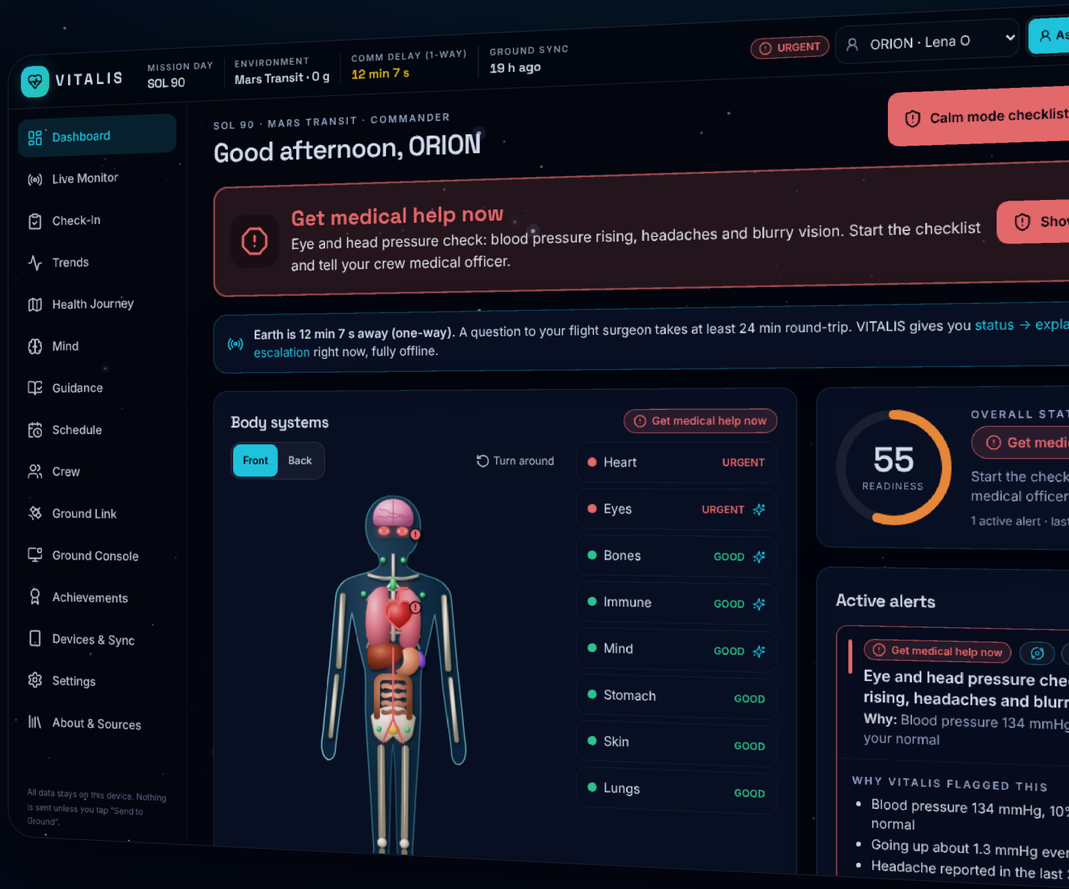
VITALIS

Your health, your call, even when Earth is 20 minutes away.

Astronaut Health Command Center: an offline-first app that helps astronauts check their own health, spot risky trends early and know exactly what to do.



Team **LUNOVA CLOUDER** · 2026 NASA Space Apps Challenge · *Create Health Monitoring Software for Astronauts on Space Missions*

THE PROBLEM

On the way to Mars, help is 45 minutes away.

4–22 min

one-way signal delay

A question to a flight surgeon and the answer can take up to 45 minutes. On the ISS it takes seconds.

0

resupply, quick way home

NASA counts **Distance from Earth** among the five hazards of human spaceflight: crews must act on their own.

"BP 133/82"

raw numbers don't help

A non-doctor needs to know: Is that bad? Why? What do I do now? When do I get help?

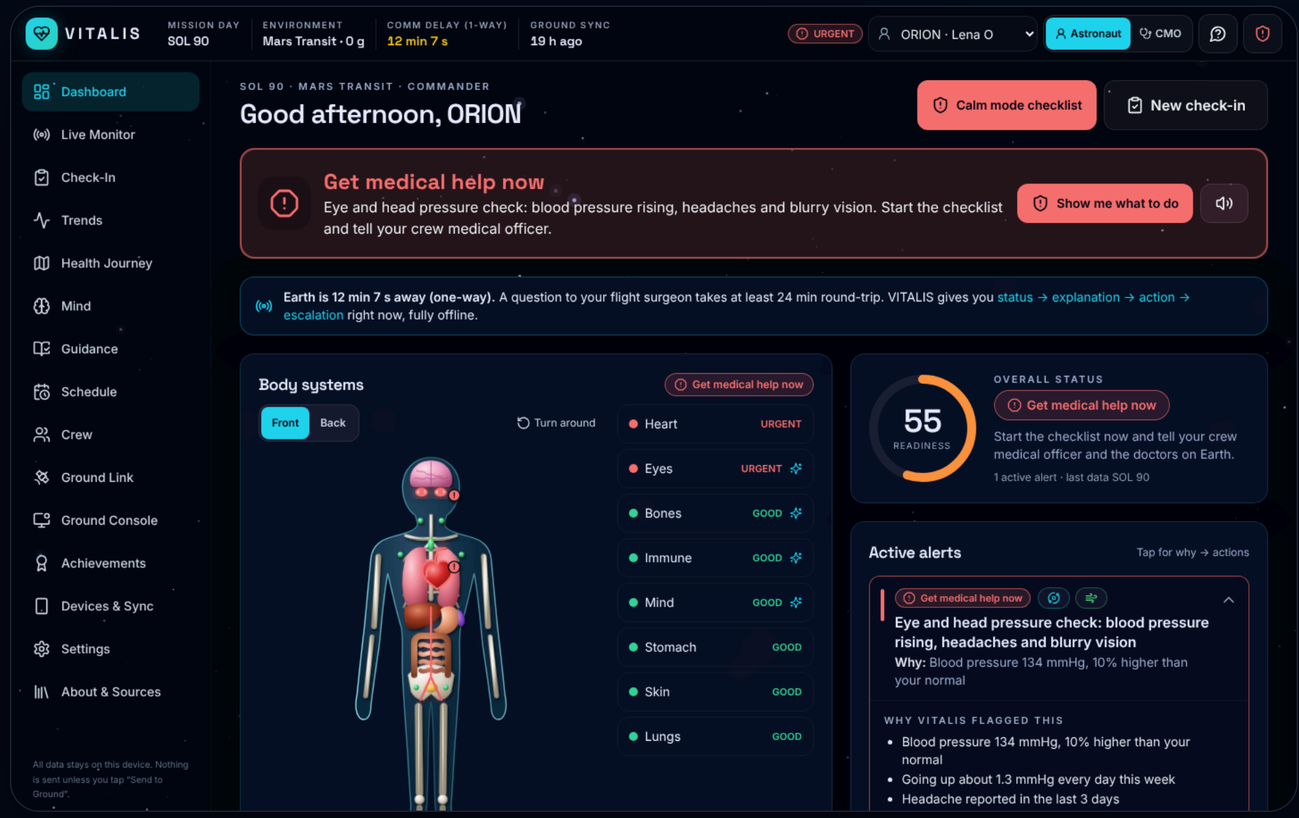
OUR SOLUTION

Every reading becomes a clear next step.

① **Status** How am I doing?
② **Why** What VITALIS noticed
③ **What to do** A tickable checklist
④ **When to get help** Timed to EarthTimed to the Earth delayapos;s delay

- All good
- Keep an eye on it
- Take action today
- Get medical help now

Plain words, sound and vibration alarms, and an emergency "calm mode" checklist. Decision support, never a diagnosis.



HOW IT DECIDES

Transparent rules that know what is normal for you.

1 · Safe limits

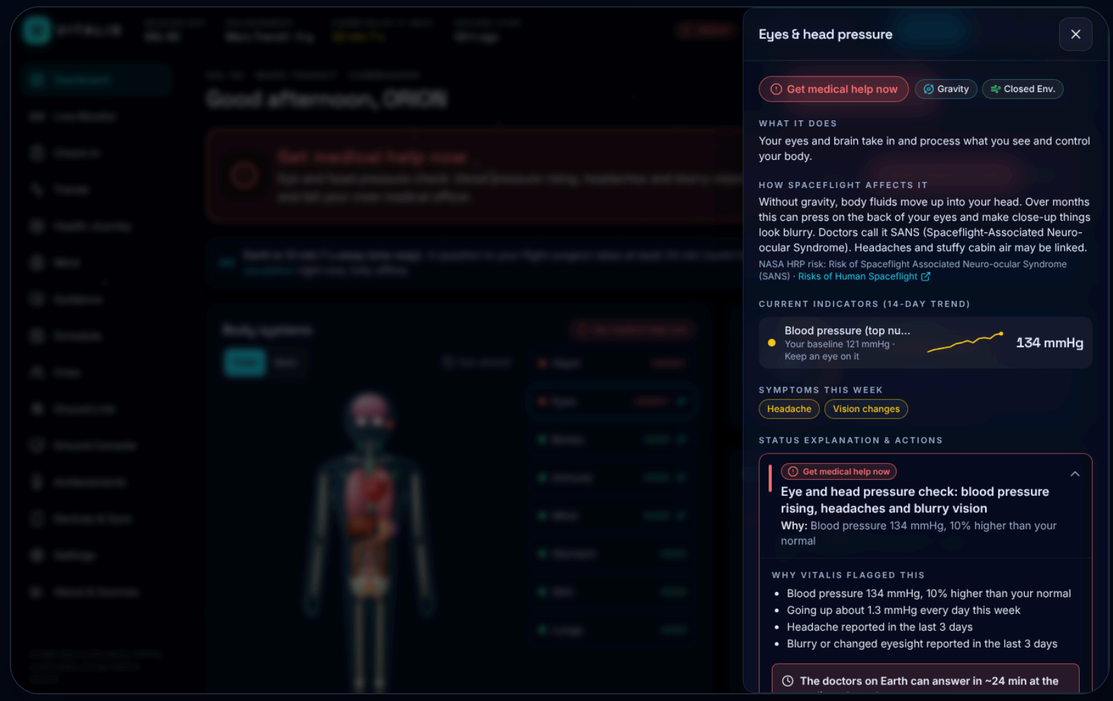
Every limit cites its source (NASA standards, clinical guidelines) and adapts to the mission, e.g. radiation for Mars vs the ISS.

2 · Your personal baseline

A 14-day average for each person catches "normal on paper, abnormal for you", and 3-day trends.

3 · 11 combination rules

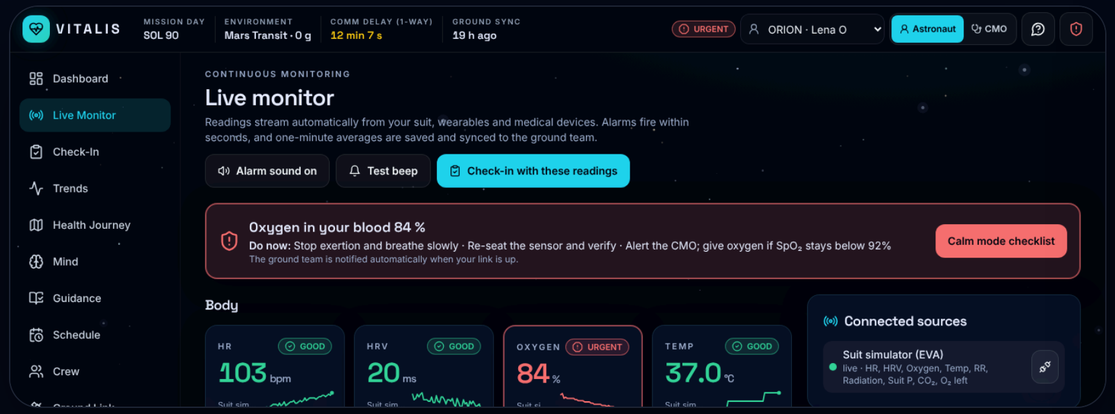
Eye & head pressure (SANS), solar storm, mood & isolation, bone loss, sleep debt, infection, kidney stones and more, each linked to a NASA risk.



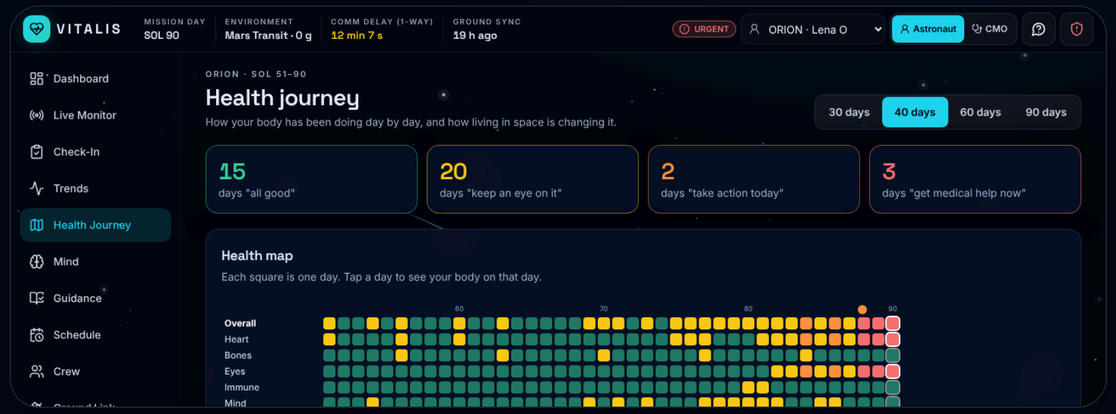
Tap an organ: why it was flagged, what to do, and "the doctors on Earth can answer in ~24 min at the earliest, start these steps now".

BUILT FOR EVERYDAY LIFE IN SPACE

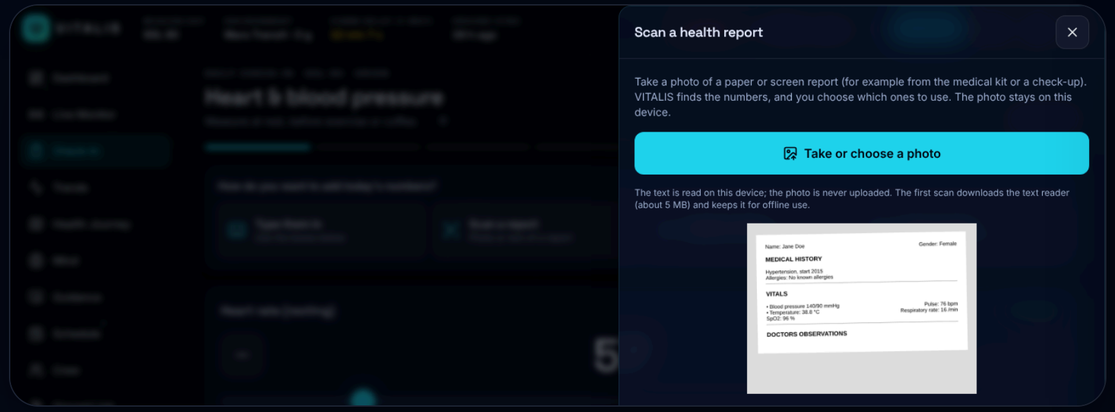
Measure, watch, understand, stay connected.



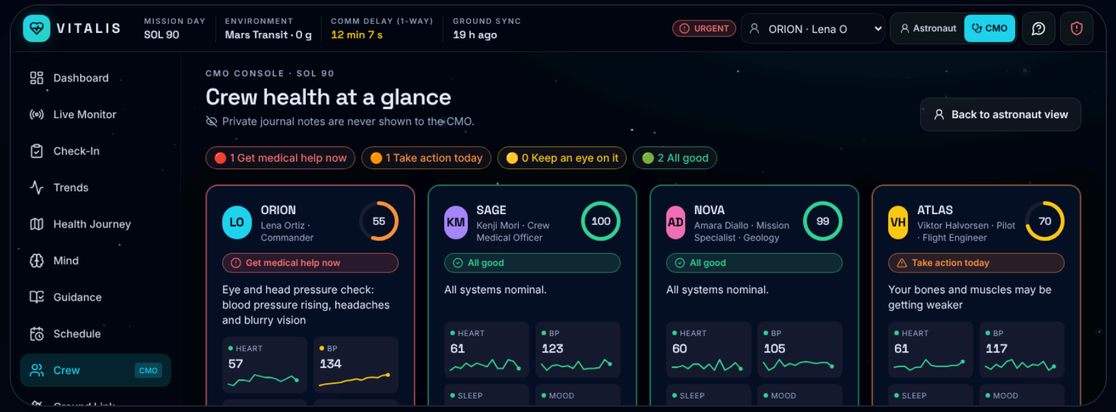
Live Monitor: suit, watch & Bluetooth devices with instant alarms



Health Journey: how space changes your body, day by day



Scan a report: numbers read from a photo, on the device



Crew & Mission Control: whole crew at a glance, alerts, messages

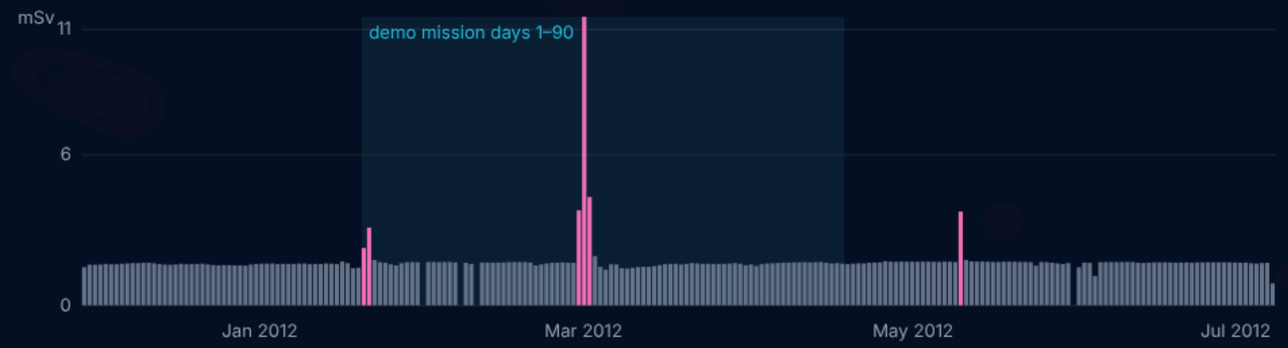
MEASURED BY NASA'S CURIOSITY ROVER ON ITS WAY TO MARS

The demo's radiation is real NASA data.

Real NASA data: radiation on the way to Mars

NASA Planetary Data System

Measured every few minutes by the Radiation Assessment Detector on NASA's Curiosity rover during its 218-day cruise (Dec 6, 2011 to Jul 14, 2012): 15,358 readings, 6 solar-storm days. The demo mission's radiation uses the shaded 90 days.



Mar 8, 2012: 11.51 mSv (absorbed 6.99 mGy, 87 readings) · solar storm

Solar storm days. Absorbed dose is measured; the mSv value is an estimate of the dose to a crew member (method in data/nasa/README.md). Data set MSL-M-RAD-3-RDR-V1.0, NASA Planetary Data System (PPI node). Hassler et al. / Zeitlin et al., Science 2013.

15,358
readings over 218 days of the cruise to Mars

8 Mar 2012
real solar storm = demo day 42 → shelter alert

All five challenge resources, used

NASA 5 Hazards tag every reading, alert and guide

NASA HRP Risks map each body area to its risk

NASA HRP Research shapes exercise, sleep and eye guides

CSA Radiation Monitoring: dosimeter check-ins, storm shelter, dose panel

CSA Science in Space: benefits on Earth

NASA PDS data set MSL-M-RAD-3-RDR-V1.0 · every source listed in DATA_SOURCES.md

WORKS WITH NO INTERNET · HELPS ON EARTH TOO

Saved on the device first. Sent to Earth when a link exists.

Fully offline

Installs as an app on phone, tablet and laptop. All checks run on the device; uploads wait in a queue and send themselves when a link returns, like NASA's store-and-forward networking.

Benefits on Earth

Remote clinics, polar stations, ships and disaster zones: anywhere with no doctor nearby and a slow or missing connection.

Built to be trusted

114 unit tests, 21 database security checks, private journal only you can see, every threshold sourced, accessible design.

Try it now, no sign-up:

vitalis-eta-lac.vercel.app/?demo=1&tour=1

Code: **github.com/QURSAM-70179225/vitalis**

Decision support, not diagnosis. Not endorsed by NASA or CSA. Prepared with SI assistance; screenshots are real captures of the app.